

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Case Study on Memory Forensics and Illegal File Transfer

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Course Title: Computer Forensics Case Study

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Computer Forensics Case Studies

1. EXECUTIVE OVERVIEW

This formal technical report details the offline volatile memory forensic examination of a physical RAM dump (`memdumpWin7.mem`, 1 GB) acquired from a Windows 7 system associated with suspected illegal document exfiltration. The objective was to reconstruct an evidence-based account of the system state at the time of capture: running user applications, network artifacts, command prompt history, removable USB media and drive-letter mapping, and File Explorer navigation artifacts (ShellBags).

Analysis with Volatility Framework 2.6.1 reconstructed a connected workflow centred on the interactive user account `IEUser`, with the memory image captured at **2019-01-06 15:09:06 UTC**. The observed activity includes:

1. **Remote document request:** the string `http://192.168.56.5/secret_file.docx` is present in Google Chrome memory, together with Omnibox search records for the same address (Figure A3e). A later command-prompt session changes into the `Downloads` directory and copies this file, which indicates the file was retrieved to that folder.
2. **Removable media connection:** a USB mass storage device (`Disk&Ven_General&Prod_UDisk&Rev_5.00`) was enumerated under `USBSTOR` at 15:02:54 UTC, and a volume mapped to drive letter **F:** appears under the user's `MountPoints2` key, last written 15:03:07 UTC.
3. **Command-line copy:** the command history of `cmd.exe` (PID 3996, hosted by `conhost.exe` PID 3988) contains, in order, `ipconfig`, `cd Downloads` and `copy secret_file.docx F:`.

Forensic Summary & Confidence Note: Memory artifacts link the active profile `IEUser` to the browser request for `secret_file.docx` and to a command-line copy of that file to removable drive F:. Confidence in profile-level activity is **Moderate to High**. Volatile memory records software and profile activity rather than human identity, and records that a command was entered rather than that the write completed. Memory evidence alone therefore cannot rule out unattended workstation use or credential sharing, nor confirm that the file reached the drive, without physical examination of drive F:.

EVIDENCE INVENTORY

To maintain chain of custody, a cryptographic SHA-256 digest of the memory image is computed before analysis, and all analysis is performed on a working duplicate inside a Kali Linux forensic environment.

Evidence ID	File Path / Type	SHA-256 Digest	Description
EVID-01	`memdumpWin7.mem` Raw memory dump	[PASTE SHA-256 FROM YOUR sha256sum OUTPUT]	Primary raw physical memory image (1 GB) from the Windows 7 target system.
EVID-02	`working_memdumpWin7.mem` Working copy	[PASTE SHA-256 FROM YOUR sha256sum OUTPUT] (must match EVID-01)	Bit-stream working duplicate used for Volatility analysis.
EVID-03	`process_1.txt` Extracted artifact text	N/A (derived text file)	Output of `strings` filtered for `192.168.56.5`, containing the requested URL.

2. TECHNICAL LAB EXECUTION & EVIDENCE ANALYSIS

Phase 1: Environment Initialization & Forensic Preservation

A working directory was created under `/home/garba/Downloads/volatility` on a Kali Linux virtual machine. The original image `memdumpWin7.mem` was made read-only with `chmod 440` and a working duplicate was created. SHA-256 digests of both files were calculated to confirm they are identical.

```
cd ~/Downloads/volatility
chmod 440 ../memdumpWin7.mem
cp ../memdumpWin7.mem working_memdumpWin7.mem
sha256sum ../memdumpWin7.mem working_memdumpWin7.mem
```

Figure A1: Terminal verification of the primary evidence file `memdumpWin7.mem` and its working copy, confirming write protection and identical SHA-256 digests.

[INSERT SCREENSHOT: sha256sum output for memdumpWin7.mem and working_memdumpWin7.mem]

Phase 2: System Profile & User Account Identification

The working image was examined with Volatility Framework 2.6.1 (`python2 vol.py`). `imageinfo` suggested several Windows 7 SP1 x86 profiles, and `Win7SP1x86` was used for all later plugins. `hivelist` located the registry hives, and `printkey` was used on `ProfileList` and `Volatile Environment` to identify the logged-in account.

```
python2 vol.py -f ../memdumpWin7.mem imageinfo
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 hivelist
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 printkey -K
"Microsoft\Windows NT\CurrentVersion\ProfileList"
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 printkey -K "Volatile
Environment"
```

Profile Field	Value	Forensic Function
Selected Profile	`Win7SP1x86` (PAE, 1 processor)	Sets the kernel data structure definitions used by all plugins.
Image Capture Time	2019-01-06 15:09:06 UTC (07:09:06 local, UTC-08:00)	Upper time boundary for all volatile evidence.
KDBG / DTB	`0x82972c68` / `0x185000`	Kernel debugger block and directory table base found by Volatility.
Active Username	`IEUser` (domain `IE8WIN7`)	Interactive user session, from Volatile Environment.
Profile SIDs	`S-1-5-21-1716914095-909560446-1177810406-1000` and `...-1002`	Two local user profiles are listed under ProfileList.

Figure A2a: Output of `imageinfo` showing suggested profiles, KDBG address and capture time 2019-01-06 15:09:06 UTC+0000.

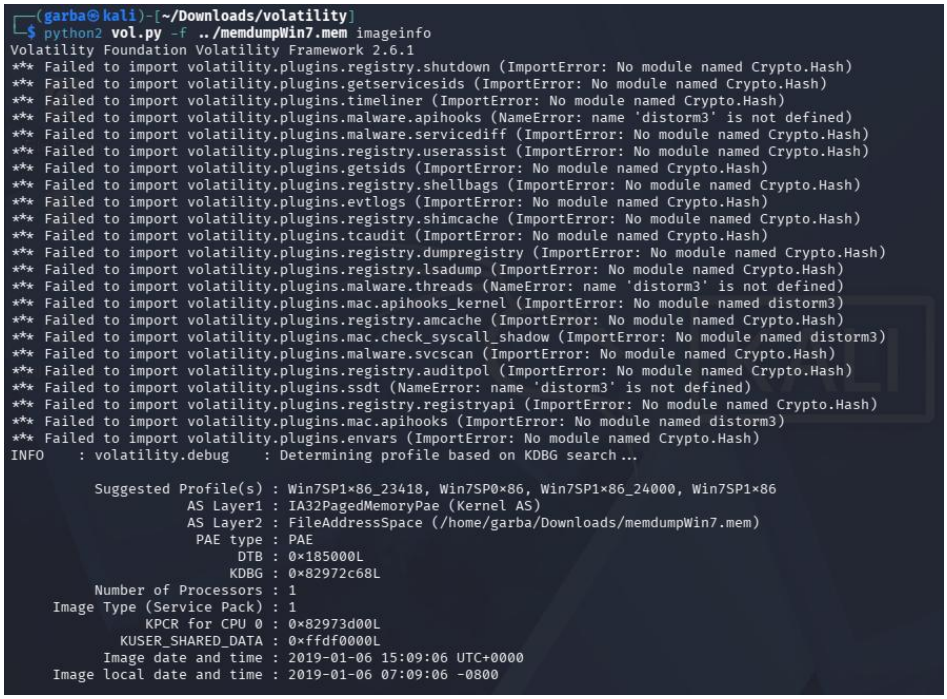
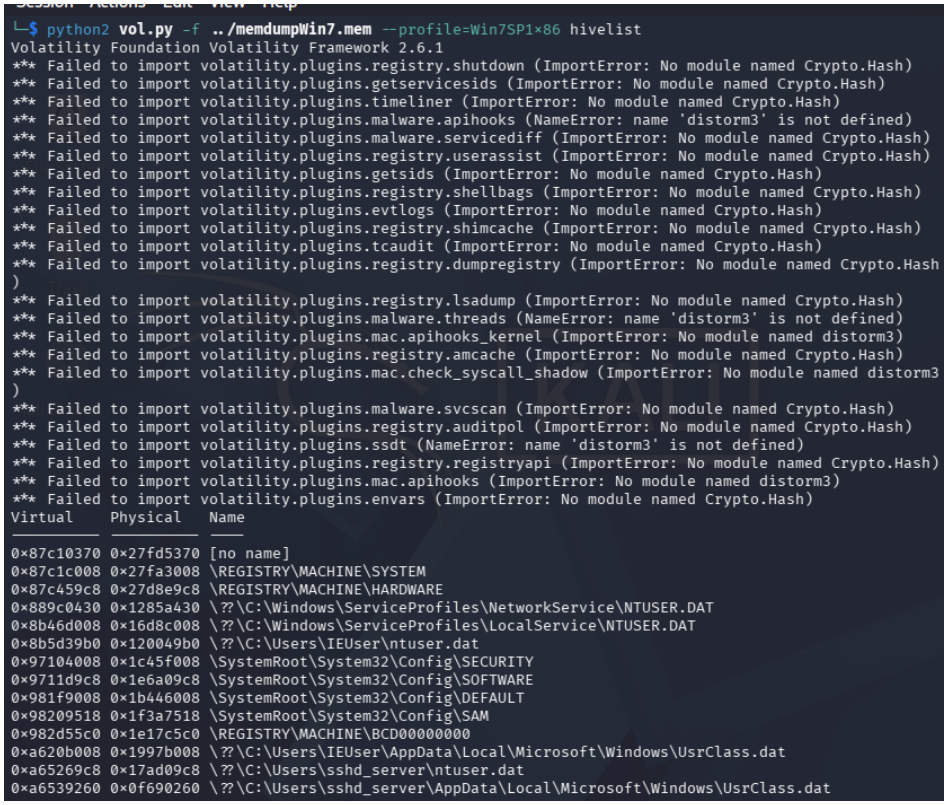


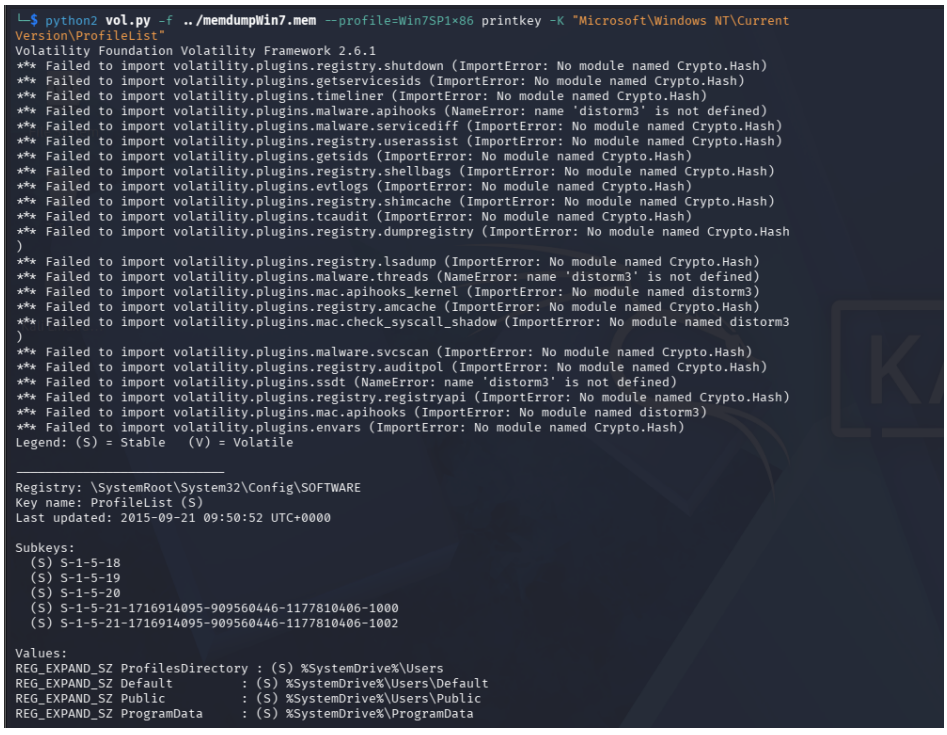
Figure A2b: Output of `hivelist` showing the SYSTEM, SOFTWARE and SAM hives and the user hives `\Users\IEUser\ntuser.dat` and `\Users\sshd_server\ntuser.dat`.



```
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 hivelist
Volatility Foundation Volatility Framework 2.6.1
** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash
)
** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3
)
** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Virtual    Physical   Name
---------- ---------- ----

0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
```

Figure A2c: Output of `printkey` on `ProfileList` listing the profile SIDs, including `S-1-5-21-1716914095-909560446-1177810406-1000`.



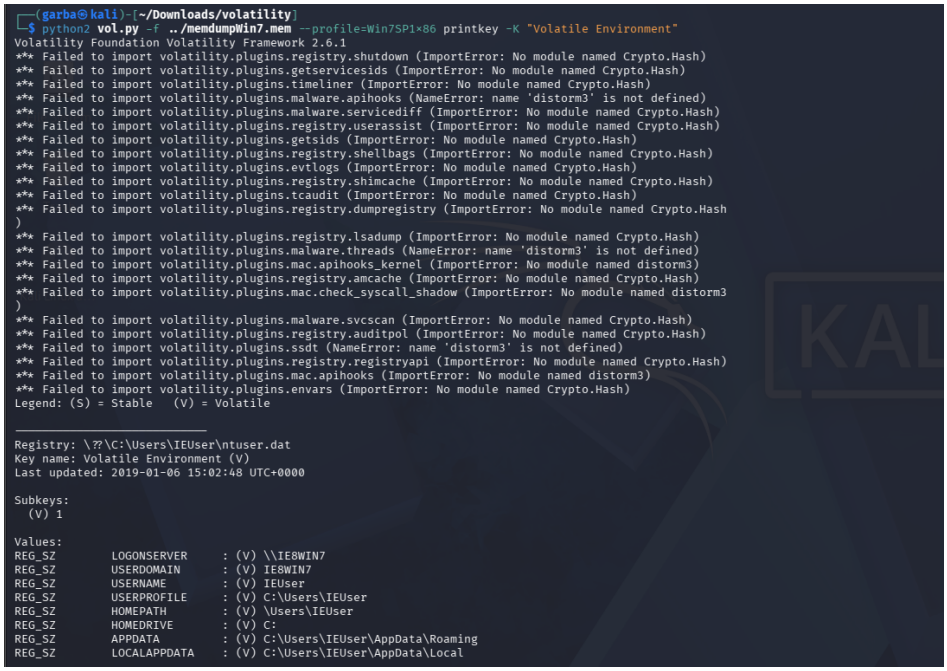
```
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 printkey -K "Microsoft\Windows NT\Current
Version\ProfileList"
Volatility Foundation Volatility Framework 2.6.1
** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash
)
** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3
)
** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData
```

Figure A2d: Output of `printkey` on `Volatile Environment` in `IEUser\ntuser.dat` (last updated 15:02:48 UTC), confirming `USERNAME = IEUser`.

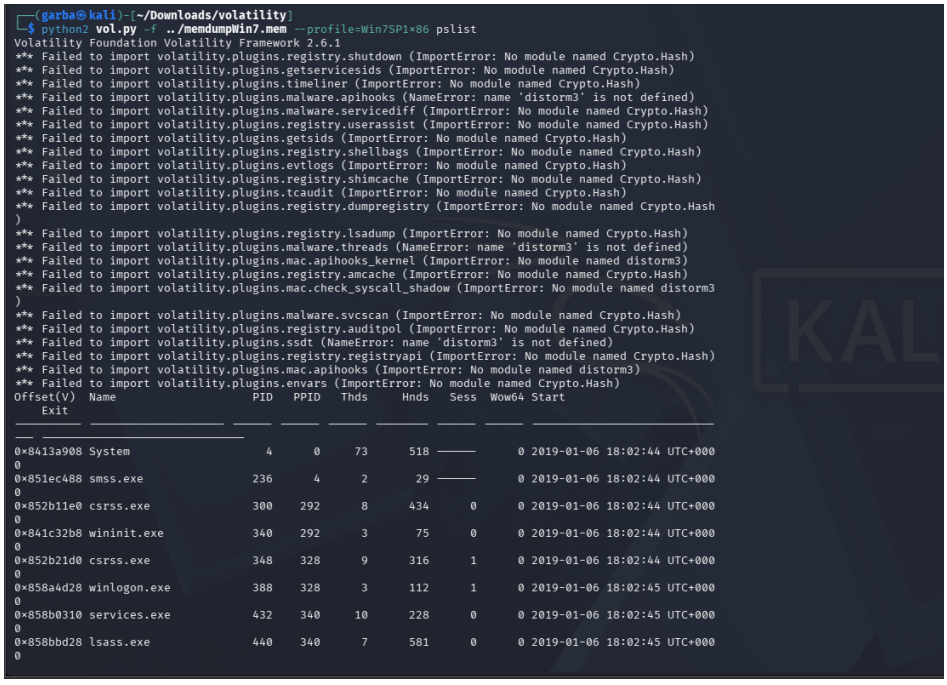


Phase 3: Network & Process Memory Artifact Extraction

Processes and sockets were enumerated with `pslist` and `netscan`. The process list shows two `chrome.exe` instances created after the session started (PIDs 2912 and 976, both children of PID 2388), an interactive `cmd.exe` (PID 3996), and `FTK Imager.exe` (PID 2596), which was started at 15:07:15 UTC and is consistent with the memory acquisition. An OpenSSH service (`sshd.exe`, PID 2016, with `cygrunsrv.exe`) was also running. `netscan` shows `chrome.exe` PID 2388 with a socket on local adapter `192.168.56.8`. A `strings` search over the image for `192.168.56.5` recovered the requested URL.

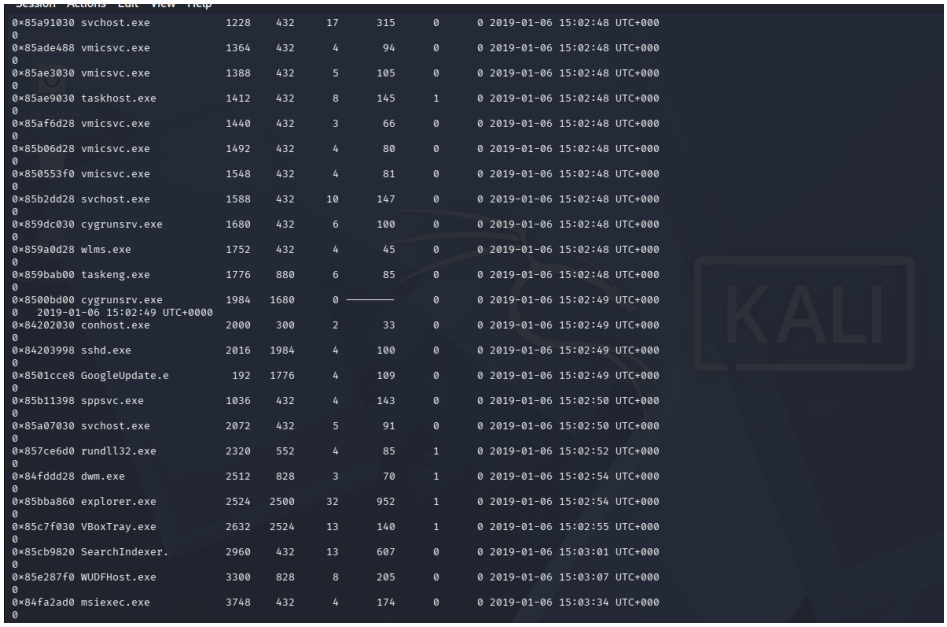
```
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 pslist
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 netscan
strings -a -t d ../memdumpWin7.mem | grep -C 3 "192.168.56.5" > process_1.txt
cat process_1.txt
```

Figure A3a: `pslist` output showing early system processes (System, smss.exe, csrss.exe, winlogon.exe, services.exe, lsass.exe).



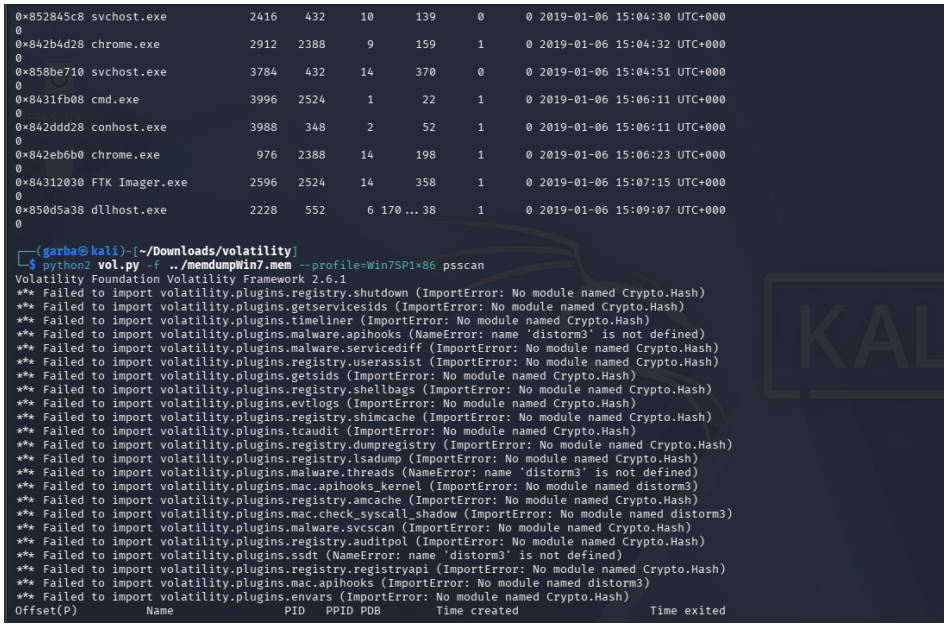
```
┌──(garba㉿kali)-[~/Downloads/volatility]
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 pslist
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash
)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3
)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start
    Exit
__ ____________________________
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+000
0
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+000
0
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+000
0
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+000
0
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+000
0
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+000
0
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+000
0
0x858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+000
0
```

Figure A3b: `pslist` output showing background services, `sshd.exe` (PID 2016), `explorer.exe` (PID 2524) and `VBoxTray.exe`.



```
0x85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+000
0
0x85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+000
0
0x85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+000
0
0x85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+000
0
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+000
0
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+000
0
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+000
0
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+000
0
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+000
0
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+000
0
0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+000
0
0x8500bd00 cygrunsrv.exe          1984   1680      0 ------      0      0 2019-01-06 15:02:49 UTC+000
0   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+000
0
0x84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+000
0
0x8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+000
0
0x85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+000
0
0x85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+000
0
0x857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+000
0
0x84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+000
0
0x85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+000
0
0x85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+000
0
0x85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+000
0
0x85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+000
0
0x84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+000
0
```

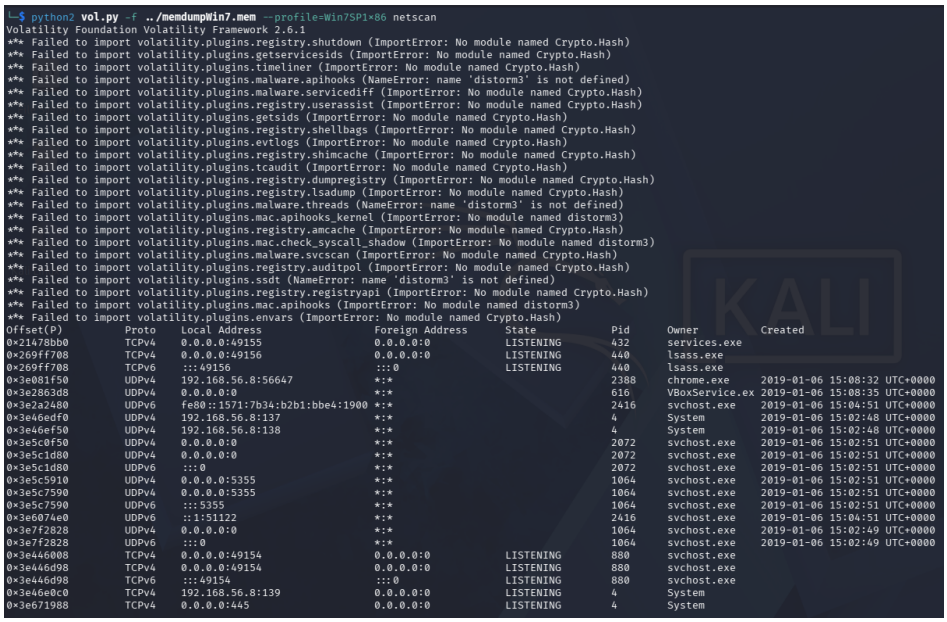
Figure A3c: `pslist` output showing `chrome.exe` (PIDs 2912 and 976, parent 2388), `cmd.exe` (PID 3996), `conhost.exe` (PID 3988) and `FTK Imager.exe` (PID 2596).



```
0x852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+000
0
0x842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+000
0
0x858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+000
0
0x8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+000
0
0x842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+000
0
0x842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+000
0
0x84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+000
0
0x850d5a38 dllhost.exe            2228    552      6 170 ... 38      1      0 2019-01-06 15:09:07 UTC+000
0

┌──(garba㉿kali)-[~/Downloads/volatility]
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 psscan
Volatility Foundation Volatility Framework 2.6.1
**~ Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Name                PID   PPID PDB        Time created                   Time exited
```

Figure A3d: `netscan` output showing `chrome.exe` (PID 2388) on local adapter `192.168.56.8` and other listening and UDP sockets.



```
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 netscan
Volatility Foundation Volatility Framework 2.6.1
**~ Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
**~ Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
**~ Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**~ Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Proto    Local Address                  Foreign Address        State           Pid      Owner          Created
0x21478bb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0              LISTENING       432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0              LISTENING       440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                   LISTENING       440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*                                    2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*                                    616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46edf0         UDPv4    192.168.56.8:137               *:*                                    4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*                                    4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*                                    2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*                                    2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*                                    2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*                                    1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*                                    1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*                                    1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*                                    1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*                                    1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING       880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING       880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                   LISTENING       880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0              LISTENING       4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0              LISTENING       4        System
```

Figure A3e: `strings` output (`process_1.txt`) showing `http://192.168.56.5/secret_file.docx` and Chrome Omnibox search strings for the same address.



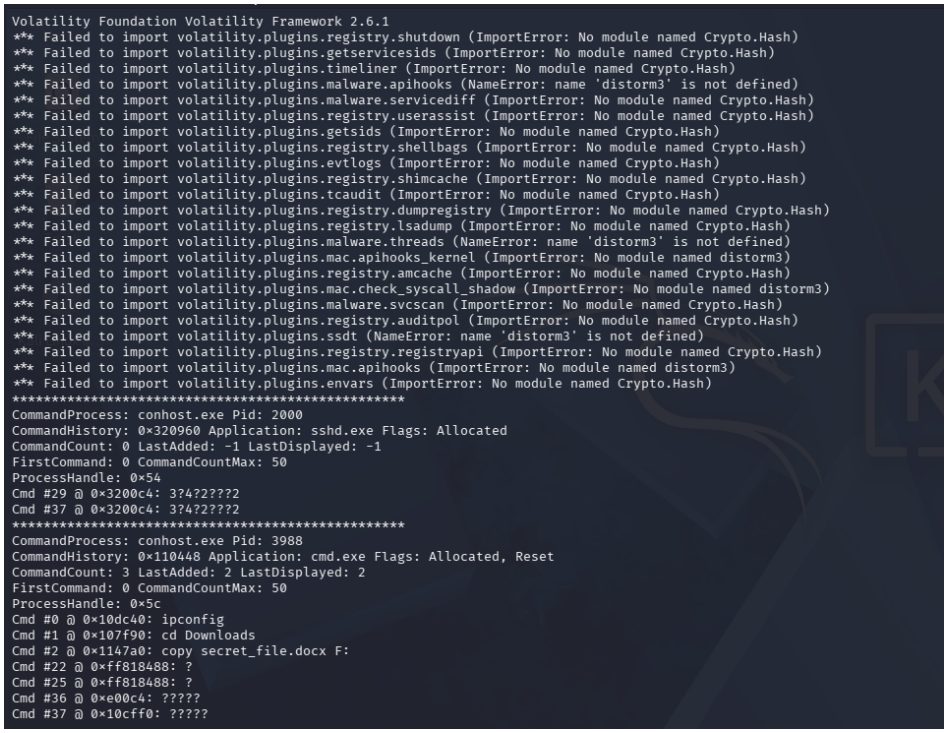
Analyst note: `strings` was run over the whole image, so these hits are not tied to a single process by the tool. The link to Chrome comes from the surrounding content (`sourceid=chrome`, Omnibox query strings). The hits also include a mistyped variant (`192.128.56.5`), consistent with the address being typed by hand. The strings show the URL was requested; they do not on their own prove the download completed.

Phase 4: Command Execution, Removable Media & ShellBags Analysis

`cmdscan` and `consoles` were used to recover command history from the console host. Registry keys `USBSTOR` and `MountPoints2` were queried to confirm the USB device and its drive-letter mapping. ShellBags extraction was attempted with the `shellbags` plugin.

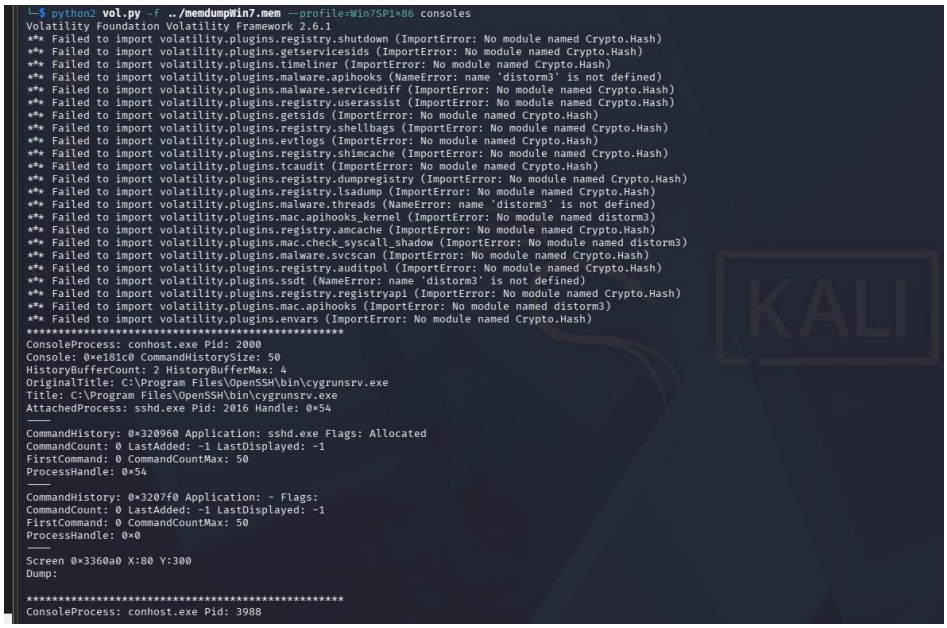
```
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 cmdscan
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 consoles
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 printkey -K
"ControlSet001\Enum\USBSTOR"
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 printkey -K
"Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 shellbags --output=html --
output-file=shellbags.html
```

Figure A4a: `cmdscan` output for `conhost.exe` (PID 3988): `ipconfig`, `cd Downloads`, `copy secret_file.docx F:`.



```
Volatility Foundation Volatility Framework 2.6.1
**** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?????2
Cmd #37 @ 0x3200c4: 3?4?????2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????
```

Figure A4b: `consoles` output listing console buffers for `conhost.exe` PID 2000 (sshd) and PID 3988.



```
└─$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86 consoles
Volatility Foundation Volatility Framework 2.6.1
**** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
**** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
**** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
---
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
---
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
---
Screen 0x3360a0 X:80 Y:300
Dump:

**************************************************
ConsoleProcess: conhost.exe Pid: 3988
```

Figure A4c: `consoles` output for PID 3988 showing the `cmd.exe` command history and the screen buffer, with prompt `C:\Users\IEUser>` and IPv4 address `192.168.56.8`.

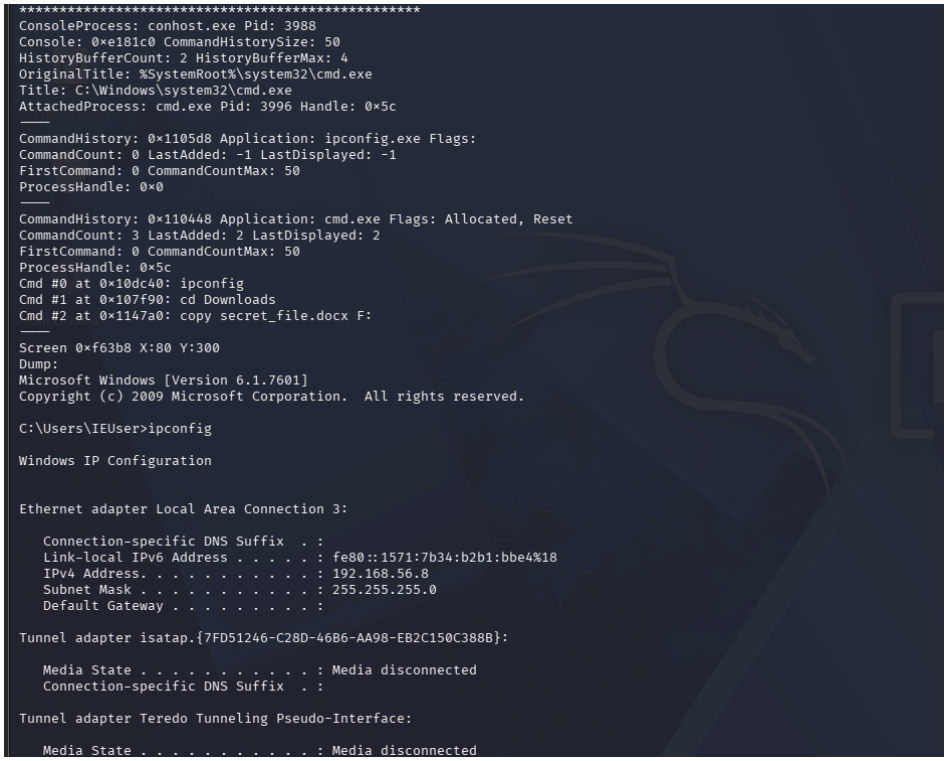


Figure A4d: `printkey` on `USBSTOR` (last updated 2019-01-06 15:02:54 UTC) showing `Disk&Ven_General&Prod_UDisk&Rev_5.00`.

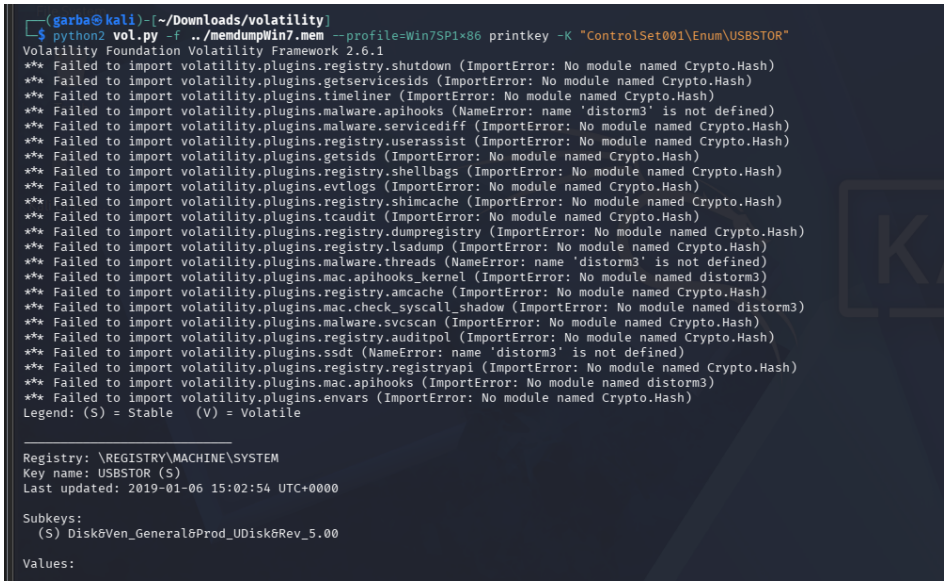


Figure A4e: `printkey` on `MountPoints2` in `IEUser\ntuser.dat` (last updated 15:03:07 UTC) showing subkey `F`.

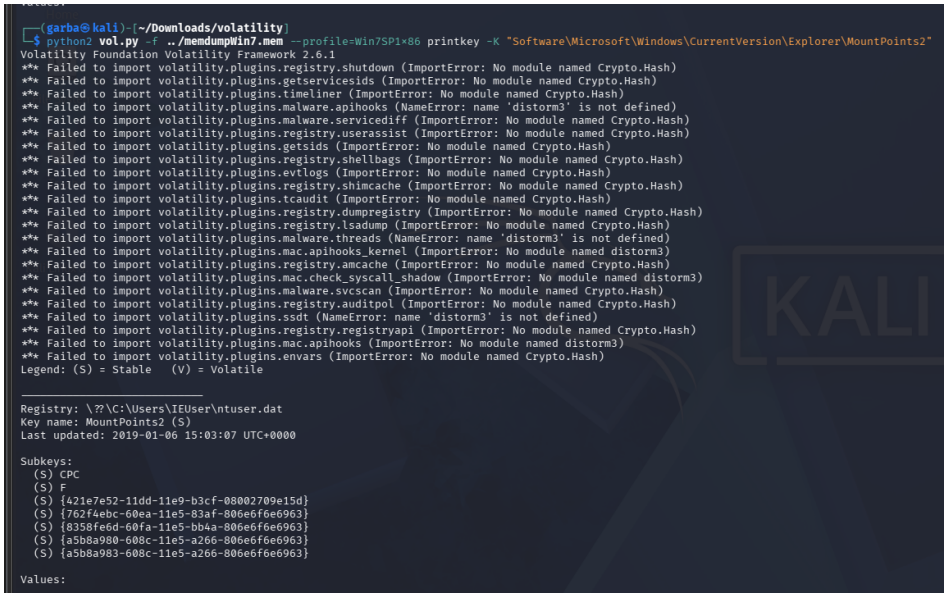
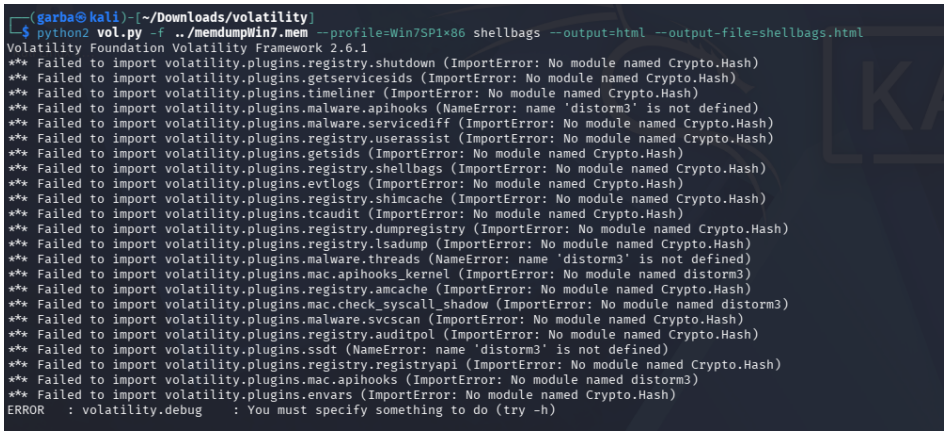


Figure A4f: Attempted `shellbags` run. The plugin did not load in this environment, so no ShellBags output was produced.



Analyst note: every Volatility run in this lab prints import errors (`No module named Crypto.Hash`, `distorm3 is not defined`). These disable the `shellbags`, `userassist`, `shimcache`, `getsids` and several other plugins, and the failed `shellbags` run is the only reason Explorer navigation history is not reported here. ShellBags findings are therefore listed as **not recovered** rather than inferred.

3. REQUIRED INVESTIGATION FINDINGS WORKSHEET

The worksheet below addresses the eight investigation tasks of the SBT-DF204 assessment brief.

No	Investigation Task	Verified Forensic Finding & Technical Citation
Q1	Acquired Evidence File & Preservation	Acquired `memdumpWin7.mem` (1 GB). SHA-256 digest: [PASTE SHA-256 FROM YOUR sha256sum OUTPUT]. The file was set read-only (`chmod 440`) and a working duplicate `working_memdumpWin7.mem` was created for Volatility 2.6.1 (Figure A1).
Q2	Relevant Profile & Image Capture Time	Profile `Win7SP1x86` (KDBG `0x82972c68`). Capture time 2019-01-06 15:09:06 UTC (07:09:06 local, UTC-08:00) (Figure A2a).
Q3	User Account & Profile Identification	Active account `IEUser` (domain `IE8WIN7`, profile `C:\Users\IEUser`) from `Volatile Environment` (Figure A2d). `ProfileList` lists user SIDs ending `-1000` and `-1002` (Figure A2c), and `hivelist` shows hives for `IEUser` and `sshd_server` (Figure A2b). The SID `S-1-5-21-1716914095-909560446-1177810406-1000` is attributed to `IEUser`; the `ProfileImagePath` value that would confirm this mapping is not visible in Figure A2c.
Q4	Process & Network Artifact Extraction	`chrome.exe` PID 2388 (parent of PIDs 2912 and 976) holds a socket on `192.168.56.8` (Figures A3c, A3d). The URL `http://192.168.56.5/secret_file.docx` and matching Omnibox search strings are present in memory (Figure A3e). No download timestamp could be recovered from the screenshots.
Q5	Command Prompt History Analysis	`cmdscan` and `consoles` recovered the history of `cmd.exe` PID 3996 (`conhost.exe` PID 3988): `ipconfig` → `cd Downloads` → `copy secret_file.docx F:`. The prompt `C:\Users\IEUser>` confirms the session account (Figures A4a, A4b, A4c). The console output of the copy command is not visible, so its success is unconfirmed.
Q6	Removable Media & Drive Letter Mapping	USB device `Disk&Ven_General&Prod_UDisk&Rev_5.00` enumerated under USBSTOR, key last written 15:02:54 UTC (Figure A4d). Drive letter F present under `MountPoints2`, key last written 15:03:07 UTC (Figure A4e).
Q7	File Explorer Navigation (ShellBags)	Not recovered. The `shellbags` plugin failed to load (missing `Crypto.Hash` module), so no ShellBags timestamps are reported (Figure A4f). Indirect support for use of the `Downloads` folder comes from the `cd Downloads` command in Q5.
Q8	Defensible Attribution & Conclusion	The `IEUser` profile requested `secret_file.docx` from `192.168.56.5` and entered a command to copy it to drive F:. Confidence: Moderate-to-High (profile level) / Low-to-Moderate (human level). Limitations: no physical or keyboard logs, drive F: not examined, ShellBags not recovered, and RAM volatility.

4. CHRONOLOGICAL FORENSIC TIMELINE

The timeline lists only events with a timestamp visible in the evidence. All times are UTC on 2019-01-06. The copy command has no recorded time; the process creation times place it between 15:06:11 and the capture at 15:09:06.

Time (UTC)	Record	Artifact / Source	Category	Observed Event
15:02:48	REG #1	Registry `Volatile Environment` (IEUser)	Session	Key last written; user session environment established (`IEUser`, `IE8WIN7`).
15:02:49	PRC #1	`pslist` / `sshd.exe` PID 2016	Service	OpenSSH service started alongside `cygrunsrv.exe`.
15:02:54	USB #1	Registry USBSTOR	Hardware	USB mass storage device enumerated by the PnP manager.
15:03:07	MNT #1	Registry `MountPoints2`	Volume mount	Subkey F present; volume mapped to drive letter F:.
15:04:32	PRC #2	`pslist` / `chrome.exe` PID 2912	Browser	Chrome child process started (parent PID 2388).
15:06:11	PRC #3	`pslist` / `cmd.exe` PID 3996, `conhost.exe` PID 3988	Shell	Command prompt session started.
After 15:06:11	CMD #1	`cmdscan` / `consoles`	Command	`ipconfig`, `cd Downloads`, `copy secret_file.docx F:` entered (no per-command time).
15:06:23	PRC #4	`pslist` / `chrome.exe` PID 976	Browser	Further Chrome child process started.
15:07:15	PRC #5	`pslist` / FTK `Imager.exe` PID 2596	Acquisition	Imaging tool started.
15:08:32	NET #1	`netscan` / `chrome.exe` PID 2388	Network	UDP socket created on `192.168.56.8:56647`.
15:09:06	MEM #1	`imageinfo` (KDBG)	Capture	Memory image captured; upper boundary of the evidence.

5. ATTRIBUTION ASSESSMENT & TECHNICAL LIMITATIONS

Observed Facts vs. Technical Inferences

- **Direct Forensic Observations:** the string `http://192.168.56.5/secret_file.docx` and related Omnibox searches are present in Chrome-related memory. `cmdscan` and `consoles` show `copy secret_file.docx F:` entered in `cmd.exe` PID 3996 under the `IEUser` prompt. Registry keys show a USB device and a drive-letter F mapping for the `IEUser` profile.
- **Reasonable Inferences:** the operator of the `IEUser` account retrieved `secret_file.docx` to the `Downloads` folder and intentionally attempted to copy it to an external USB drive.

Technical Limitations & Alternative Hypotheses

1. **User profile vs. physical identity:** memory records activity under the `IEUser` profile. It cannot show who was at the keyboard without secondary evidence such as CCTV, access or biometric logs.
2. **Shared or unlocked terminal:** another person could have used an unattended, logged-in workstation.
3. **Remote access:** an OpenSSH server (`sshd.exe`, with an `sshd_server` account) was running, so remote access to the machine cannot be excluded without reviewing SSH logs.
4. **Execution vs. disk write:** the command history shows the command was entered, not that the copy succeeded. Without a bit-stream image of drive F: the write and the destination file hash cannot be verified.
5. **Download not time-stamped:** the URL strings show a request for the file, but no download time or completion record was recovered.
6. **Plugin failures:** `shellbags`, `userassist`, `shimcache` and other plugins failed to load, so those artifact sources were not examined.
7. **Inconsistent early-boot timestamps:** the first `pslist` page shows start times of 18:02:44–18:02:45 for early system processes, later than the 15:09:06 capture time. These are treated as unreliable and are not used in the timeline.

6. EVIDENCE LOG APPENDIX

Item ID	Source Record / Artifact	Forensic Finding & Technical Observation	Screenshot Ref
E01	File system / `sha256sum`	SHA-256 digest of `memdumpWin7.mem`: [PASTE SHA-256 FROM YOUR sha256sum OUTPUT]	Figure A1
E02	Volatility `imageinfo`	Profile `Win7SP1x86`; capture time 2019-01-06 15:09:06 UTC	Figure A2a (Screenshot 486)
E03	Volatility `hivelist`, `printkey`	Active user `IEUser`; profile SIDs ending -1000 and -1002	Figures A2b, A2c, A2d (Screenshots 518, 519, 520)
E04	Volatility `pslist`, `netscan`	`chrome.exe`, `cmd.exe`, `conhost.exe`, FTK `Imager.exe`, `sshd.exe`; Chrome socket on 192.168.56.8	Figures A3a–A3d (Screenshots 521, 522, 523, 524)
E05	Linux `strings` / `process_1.txt`	URL `http://192.168.56.5/secret_file.docx` in memory	Figure A3e (Screenshot 528)

Item ID	Source Record / Artifact	Forensic Finding & Technical Observation	Screenshot Ref
E06	Volatility `cmdscan`, `consoles`	Recovered command `copy secret_file.docx F:` in `cmd.exe` PID 3996	Figures A4a, A4b, A4c (Screenshots 529, 530, 531)
E07	Volatility `printkey` (USBSTOR, `MountPoints2`)	USB device `Disk&Ven_General&Prod_UDisk&Rev_5.00`; volume mapped to F:	Figures A4d, A4e (Screenshots 532, 533)
E08	Volatility `shellbags`	Plugin failed to load; no output obtained	Figure A4f (Screenshot 534)

7. ACADEMIC INTEGRITY & AUTHORIZATION STATEMENT

I confirm that I completed this digital forensics case study independently in an authorized virtual environment using the supplied historical memory forensics scenario dataset. I have accurately documented my own Volatility commands, record citations, evidence log entries, cryptographic hashes, and forensic observations without fabrication.

Investigator Name: Ahmad Zubairu Garba

Student ID: 2025/FWSD/11463

Course Code: SBT-DF204

Date: October 17, 2026